ResearchPaper Lens — Forensic & Analytical Report

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1. Document Integrity & Cryptographic Evidence

Metric / Digest	Value
Filename:	sample_05_academic_paper.pdf
File Size:	2,594 bytes
Pages / Words / Sentences:	1 pages | 163 words | 18 sentences
References Section:	None detected
SHA-256 Digest:	`7b5b89654e23a22c201c460b687016ad0e6a1b6625ea1f4cf48d4d350e1271d8`
SHA-1 Digest:	`581a88c7ba9d89049c2331645cb53d32cedab1d1`
MD5 Digest:	`3ad7b3760a10991e614264338bfbd29a`

2. Explainable Review Cues & Limitations

- Extractive Faithfulness: TextRank extracts verbatim sentences directly from source text (0% hallucination risk).
- Graph Centrality: Vertex importance is determined via PageRank (d=0.85, max_iter=100) over TF-IDF cosine similarities.
- Chronological Ordering: Extracted sentences are strictly restored to their original document sequence.
- Scanned PDF Limitation: Low-text heuristic (< 100 chars/page) flags image-only PDFs where embedded text is minimal.
- Lexical Search Boundary: Passage retrieval uses lexical TF-IDF matching and does not perform neural semantic re-ranking.
- Privacy by Default: Sensitive personal identifiers, emails, and credentials are automatically masked.

3. TextRank Extractive Summary (40 words, 24.5% compression)

Extracted 3 sentences out of 18 using TF-IDF PageRank graph centrality.

A Unified Framework for TextRank Information Retrieval in Academic Literature Dr. A. In this paper, we propose a graph-theoretic approach using sentence centrality over TF-IDF cosine similarity spaces. Graph Formulation: Sentences are represented as vertices in an undirected complete graph.

4. Academic Baseline Summaries

Method	Output
Lead-k: *(First k sentences)*	A Unified Framework for TextRank Information Retrieval in Academic Literature Dr. A. Turing, Prof. J. von Neumann — Institute for Advanced Study Abstract: Automatic summarization of scientific literature is essential for accelerating research discovery. In this paper, we propose a graph-theoretic approach using sentence centrality over TF-IDF cosine similarity spaces.

Word-Frequency: *(Term count weighting)*	A Unified Framework for TextRank Information Retrieval in Academic Literature Dr. A. Graph Formulation: Sentences are represented as vertices in an undirected complete graph. Mihalcea, TextRank: Bringing Order into Text, EMNLP 2004.

5. Salient Keywords (TF-IDF Aggregation)

brin: 1.00	**references**: 1.00	**textrank**: 0.88
graph: 0.84	**retrieval**: 0.72	**academic**: 0.64
literature: 0.58	**order**: 0.42	**text**: 0.42
mihalcea: 0.42		

Confidentiality & Privacy: Redactions applied (0 sensitive patterns masked). Document evaluated locally with ResearchPaper Lens.